Seat No:_____ **Enrolment No:_____**

LOK JAGRUTI KENDRA UNIVERSITY
MCA Semester – 1 Remedial Examination Winter 2025

Subject Code: 40110201/140110201 **Date: 01/12/2025**

Subject Name: Data Structures

Duration: 02:30 Hours **Total Marks: 50**

Instructions:

1. Attempt all the questions
2. Figures on the right indicate the marks.
3. Draw the figures and give suitable examples as and where necessary.

Q - 1 Answer The Following Questions.

1. Define: Algorithm 1

2. Give two example of a primitive data type. 1

3. List Asymptotic Notations with its symbols. 1

4. Differentiate between liner and non-liner data structures. 3

5. Explain the classification of data structures with example. 4

OR

5. Define time complexity and space complexity. Why are they important in analyzing algorithms? 4

Q - 2 Answer The Following Questions.

6. Write simple queue full condition. 1

7. Write applications of stack. 1

8. Define an array. 1

9. Convert Infix expression into postfix expression using stack: 3

   (A + B) * (C - D / E)

10. Explain in detail the types of linked lists: singly, doubly, and circular linked lists with diagrams. 4

OR

10. Define: stack. Write an algorithm for push and pop operation on stack. 4

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Q - 3 Answer The Following Questions.

11. What is a leaf node? 1

12. What is an expression tree? 1

13. What are parallel edges in Graph? 1

14. Explain adjacency matrix representation of graph with example. 3

15. Construct a binary search tree from the following data and find in-order, pre-order, and post-order traversal: 4

    40, 20, 10, 30, 60, 50, 70, 65, 80

OR

15. Construct AVL tree for the following data. (show every step) 4

    30, 20, 40, 10, 25, 50, 5

Q - 4 Answer The Following Questions.

16. Define: Hash function 1

17. What is the time complexity of Bubble sort algorithm? 1

18. Define: Insertion Sort 1

19. Define Collision. Explain any two techniques for collision resolution with suitable example. 3

20. Sort following data using Radix sort. (show every step) 4

    170, 45, 75, 90, 802, 24, 2, 66

OR

20. Sort following data using Quick sort. (show every step) 4

    45, 23, 89, 12, 67, 34, 90, 11, 50, 28

Q - 5 Answer The Following Questions.

21. Queue is abstract data type (ADT). State True or False. 1

22. What is a sparse matrix? 1

23. List applications of Graph. 1

24. Define searching. Write an algorithm for Binary search. 3

25. Apply Dijikstra‘s algorithm to find shortest path on following graph with node S as the starting node. 4

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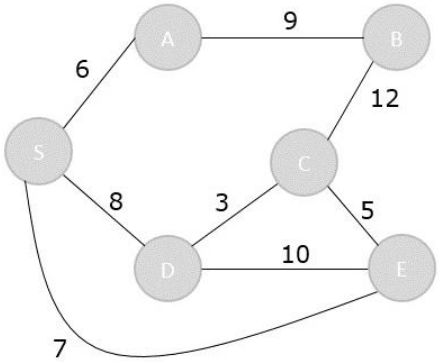
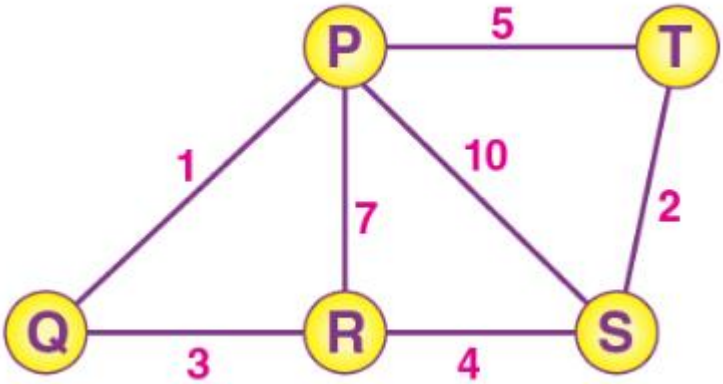


OR

25. Find minimum spanning tree of following graph using Kruskal's algorithm. 4



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